Province of the
EASTERN CAPE
EDUCATION

NATIONAL SENIOR CERTIFICATE / *NASIONALE SENIOR SERTIFIKAAT*

GRADE/*GRAAD* 10

NOVEMBER 2018

TECHNICAL SCIENCES P1/ *TEGNIESE WETENSKAPPE V1* MARKING GUIDELINE/*NASIENRIGLYN*

MARKS/*PUNTE*: 150

This marking guideline consists of 7 pages./
Hierdie nasienrglyn bestaan uit 7 bladsye.

QUESTION/*VRAAG* 1

1.1 D ✓✓
1.2 D ✓✓
1.3 A ✓✓
1.4 A ✓✓
1.5 C ✓✓
1.6 B ✓✓
1.7 D ✓✓
1.8 C ✓✓
1.9 C ✓✓
1.10 A ✓✓ (10 x 2) **[20]**

QUESTION/*VRAAG* 2

2.1 Scalar is a quantity that has magnitude only ✓✓
Skalaar is 'n hoeveelheid wat slegs grootte het. ✓✓
Vector is a quantity with both magnitude and direction ✓✓
Vektor is 'n hoeveelheid met beide grootte en rigting. ✓✓ (4)

2.2 2.2.1 2,23 ✓✓ (kg) (2)

2.2.2 2 900 ✓✓ (m) (2)

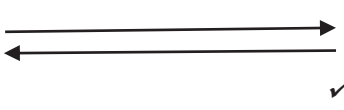
2.2.3 $2,23 \times 10^3$ ✓✓ (2)

2.3 Shortest distance between two points in a particular direction **OR**
Straight line distance from the starting point to the finishing point with direction ✓✓
Kortste afstand tussen twee punte in 'n spesifieke rigting ***OF***
Die reguitlyn tussen die beginpunt en eindpunt met rigting ✓✓ (2)

2.4 2.4.1 2,9 km ✓(or 2 900 m) (1)

2.4.2 2 km ✓(or 2 000 m) to the RIGHT / *na REGS* ✓ (2)

2.5 v = displacement/time *v = verplasing/tyd* ✓
= 2 000 ✓/1 800 ✓=1,11 $m.s^{-1}$ ✓ (4)

2.6  ✓ ✓
Resultant displacement/*Resultante verplasing* = 0 N ✓ (3)

2.7.1 No / *Nee* ✓
There are equal changes in displacement in equal time intervals ✓
Therefore velocity is constant ✓
Daar is gelyke veranderings in verplasing in gelyke tydintervalle ✓
Dus is die snelheid constant ✓ (3)

2.7.2 T= 1/F= 1/50 ✓= 0,02s ✓
Time/*tyd* = 5 x 0,02 ✓= 0,1 s ✓ (4)

2.7.3 v = displacement /time= 5 ✓x (20/100) ✓/0,1✓= 2 m.s^{-1}✓
v = verplasing/tyd = 5 ✓x (20/100) ✓/ 0,1 ✓= 2 m.s^{-1} ✓

OR

v = $\frac{Displacement\ /Verplasing}{Time/Tyd}$ ✓

= $\frac{20\ cm}{0,1}$ ✓✓

= $\frac{0,2m}{0,1}$

= 2 m.s^{-1} ✓ (4)

[33]

QUESTION/*VRAAG* 3

3.1 Single force which can produce the same effect as two or more forces ✓✓
'n Enkel krag wat dieselfde effek het as twee of meer kragte ✓✓ (2)

3.2 3.2.1 F_1 Normal/*Normaal* ✓ (1)

3.2.2 F_2 Applied force/*Toegepaste krag* ✓ (1)

3.2.3 F_3 Weight or Force of gravity/*Gewig of Swaartekrag* ✓ (1)

3.2.4 F_4 Friction/*Wrywing* ✓ (1)

3.3 Weight/*Gewig* ✓or/*of* Force of gravity/*Swaartekrag* ✓ (1)

3.4 Fg = mg ✓= 60 x 9,8 ✓= 588 N ✓ (3)

3.5 Take right as positive/*Neem regs as positief*
$F_{resultant}$ = 50 + (-15) = 35 N✓✓to the right/*na regs* ✓ (3)

OR/*OF*

Take left as positive/*Neem links as positief*
$F_{resultant}$ = 15 + (-50) = -35 N = 35 N ✓✓to the right/*na regs* ✓

3.6 3.6.1 $F_{resultant}$ = 0 N ✓ (1)

3.6.2 Equilibrant /*Ekwilibrant* ✓ (1)

3.6.3 **F** = 35N ✓to the LEFT/*na LINKS* ✓ (2)
[17]

QUESTION/*VRAAG* 4

4.1 4.1.1 Moment ✓(of a force/*van 'n krag*)/*Wringkrag* (1)

4.1.2 Cantilever / *Cantilever* ✓ (1)

4.1.3 Mechanical advantage / *Meganiese voordeel*✓ (1)

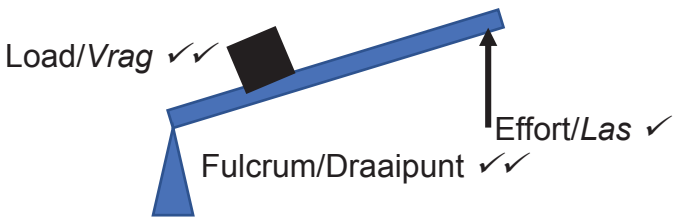
4.2 4.2.1 Sum of moments clockwise/ *Som van momente kloksgewys*
= F_1 x d_1 + F_2 x d_2 ✓
= 3 000 x 2 ✓+ 4 000 x 4 ✓
= 22 000 N.m ✓ or (22 kN)
(4)

4.2.2 $M_{ACW} = M_{CW}$ ✓ (Law of moments/*Wet van momente*)
R_B x 6 = 22 000 ✓
R_B = 3 670 N

$R_A + R_B = F_1 + F_2$ ✓(Beam is in equilibrium/*Balk is in ewewig*)

R_A + 3 670 ✓= 3 000 + 4 000 ✓
R_A = 3 330 N ✓ (6)

4.3



(6)

4.4 4.4.1 Class ONE/*Klas EEN* ✓
Load is between effort and fulcrum/*Vrag is tussen las en fulkrum* ✓ (2)

4.4.2 $F_L/F_E = d_E/d_L$ ✓

200/50 ✓= (20/100) ✓/d_L

d_L= 0,05 m ✓ (4)
[25]

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QUESTION/*VRAAG* 5

5.1.1 Energy a body has due to its position above the ground. ✓✓ (2)
Energie van 'n liggaam as gevolg van sy posisie bokant die grond ✓✓

5.1.2 Sum of kinetic energy and potential energy ✓✓ (2)
Som van die kinetiese en potensiële energie ✓✓

5.2.1 $E_{k\ at\ P} = ½\ mv^2$ ✓
$= ½ \times 60 \times 4^2$ ✓ (3)
$= 480$ J ✓

5.2.2 $U_{at\ Q} = mgh$ ✓
$= 60 \times 9{,}8 \times 3$ ✓ (3)
$= 1\ 764$ J ✓

5.3 $U_{at\ P} = mgh$
$2\ 469$ ✓ $= 60 \times 9{,}8 \times h$ ✓
4,2 m ✓ $= h$ *Accept/Aanvaar* 4,199 m (3)

5.4 $E_{k\ at\ Q} = ½\ mv^2$
705,6 ✓ $= ½\ (60)$ ✓v^2
$v = 4{,}85\ m.s^{-1}$ ✓ (3)
[16]

QUESTION/*VRAAG* 6

6.1 REMOVED /*VERWYDER* ✓ (1)

6.2 $n_e = \Delta Q/e$ ✓
$= 6 \times 10^{-6}/1{,}6 \times 10^{-19}$ ✓
$= 3{,}75 \times 10^{-19}$ ✓ e^- (3)

6.3 To prevent charge from leaking ✓✓ (2)
Om te verhoed dat lading uitlek ✓✓

6.4 6.4.1 Total charge of an isolated system remains constant ✓✓ (2)
Totale lading van 'n geïsoleerde system bly constant. ✓✓

6.4.2 From **R** to **P** / *Van **R** na **P*** ✓ (1)

6.4.3 $Q_{total\ before}$ / $Q_{totaal\ voor} = (+6 \times 10^{-6} + 0)$
$Q_{total\ after} = Q_{total\ before}$ / $Q_{totaal\ na} = Q_{totaal\ voor}$
$Q_R = +6 \times 10^{-6} / 2$ ✓✓ $= +3 \times 10^{-6}$ C ✓ (3)
[12]

QUESTION/*VRAAG* 7

7.1 Rate of flow of charge/ *Tempo van vloei van lading* ✓✓ (2)

7.2 7.2.1 CIRCUIT/*STROOMBAAN* 2 ✓ (1)

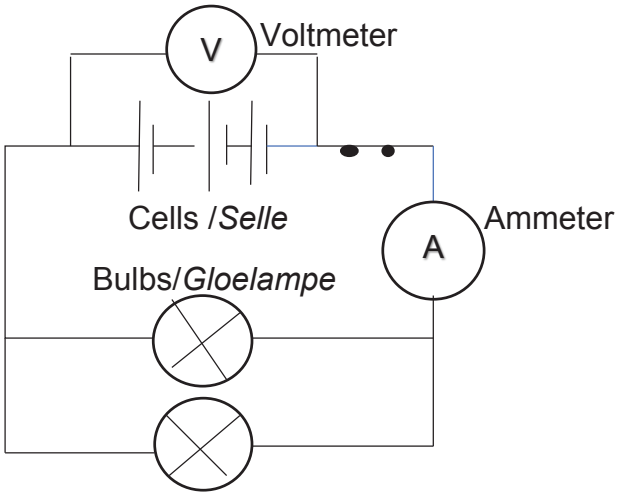
7.2.2 CIRCUIT/*STROOMBAAN* 1 ✓ (1)

7.3 **Y** to/*na* **X** ✓ (1)

7.4 7.4.1 EMF = 12 V ✓ (1)

7.4.2 $Rp= R_1R_2/R_1+R_2$ ✓ = 12 x 12 /(12 + 12) ✓ = 6 Ω ✓ (3)

7.5



Marking criteria/*Merk kriteria*
Three cells / *Drie selle* ✓
Voltmeter ✓
Ammeter ✓
Bulbs in parallel/*Gloeilampe in parallel* ✓
Switch/*Skakelaar* ✓

(5)

[14]

QUESTION/*VRAAG* 8

8.1	8.1.1	Opposition to flow of electric current ✓✓ *Opposisie teen die vloei van elektriese stroom* ✓✓	(2)
	8.1.2	Length of conductor/*Lengte van die geleier* ✓	(1)
	8.1.3	Temperature/*Temperatuur* ✓ Thickness (Cross sectional area)/ *Dikte (Deursnit-area)* ✓ Type of conductor/*Tipe geleier* ✓	(3)
8.2	8.2.1	The longer the conductor the higher the resistance OR The shorter the conductor the lower the resistance OR Resistance is directly proportional to the length of the conductor ✓✓ *Hoe langer die geleier, hoe hoër is die weerstand* ***OF*** *Hoe korter die geleier, hoe laer is die weerstand* ***OF*** *Weerstand is direk eweredig aan die lengte van die geleier* ✓✓	(2)
	8.2.2	9 √√ (Ω)	(2)
	8.2.3	Gradient/Gradiënt = (18 – 9) √/(7 – 3) √ = 2,25 √	(3) **[13]**

TOTAL/*TOTAAL*: 150

Kopiereg voorbehou